

INSuRE Fall 2022

Final Report - October 23, 2022

Critical Infrastructure Attack Demonstration

Sponsor: MITRE

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1. Project Summary

1.1 Top Level Project Description

This project aims to detect the exploits used by adversaries in attacking Critical Infrastructure Systems. Our team has identified a simulated ICS network developed by Matthew Frank for Georgia Tech's CS 6263: Intro to Cyber-Physical Systems Security called the Graphical Realism Framework for Industrial Control Simulation (GRFICSv3). The network consists of four VMs (a 3D simulation, a soft Programmable Logic Controller, an Human Machine Interface, and a workstation) communicating with each other on a host-only virtual network. We aim to find and attack all the buffer overflow vulnerabilities on the network whilst maintaining a detailed document on the exploitation process, along with mapping them to the MITRE ATT&CK for ICS Matrix. We hope to leverage this project to gain an understanding on the exploits used on such systems and networks; and potentially detect and defend against such security threats in the real world systems

1.2 Steps to Ensure Project Completion

We aim to set up weekly meetings with our technical directors to discuss expected outcomes, hurdles, and results; and leverage their wisdom and guidance throughout the duration of the project. Our first goal is to get the project infrastructure setup, which might prove to be resource intensive as it involves running multiple virtual machines connected to a host only network. Our next step would be to explore each of the systems on the network and gather information on their inner workings; communications with each other; vulnerabilities etc. After establishing a good idea of this, we aim to begin our exploitation of the systems. Throughout this process we will keep a detailed document on the different phases of our exploitation journey. Lastly, we will ensure that our team consistently meets all the deadlines as well as the expectations of our technical directors.

1.3 Estimated Project Budget

There are no monetary costs as we will be utilizing virtual machines (VMs).

1.4 Timeline of Milestones

Milestones	Tentative Completion Date
Project Proposal	September 9, 2022
Project Infrastructure Setup	September 9, 2022 - September 16, 2022
Conduct Research and Explore VMs	September 16, 2022 - September 30, 2022
Phase One of Exploit - Begin Documentation and Identify Vulnerabilities	September 30, 2022 - October 20, 2022
Midterm Presentation	October 21, 2022
Phase Two of Exploit - Continue Documenting Vulnerabilities and Look Into Security Threat Detection Mechanisms	October 22, 2022 - November 11, 2022
Phase Three of Exploit - Finalize Report and Map Attack Vectors to The MITRE ATT&CK for ICS Matrix	November 11, 2022 - December 2, 2022
Final Report Presentation	December 9, 2022

1.5 Motivation

Critical Infrastructure Systems such as Chemical Power Plants, Communication Systems, Transportations Systems etc play a vital role in a nation and failure / destruction of any one of them could have a debilitating effect on security, national economy, and public health and safety. Such systems are generally lacking in security which is a huge cause of concern considering the modern digital landscape. With increasing cyberattacks targeting Industrial Control Systems (ICS) in recent years, aises a greater need for more active security research into this field to better secure such systems. The Critical Infrastructure Attack Demonstration project by MITRE is a great opportunity to introduce us to conducting security research on ICS systems.

1.6 Anticipated Issues and Challenges

Our initial challenge would be to set up a working project infrastructure due to multiple VMs being connected together, which might be too resource intensive. In the event that a single computer cannot handle running all four VMs, we will look into leveraging more computers with each running one of the four VMs. Since the initial setup uses a host only network, we might have to configure the VMs to use a bridged network to allow communication with the other VMs as they are on different devices on the network.

Although most of the team members have experience in low level code analysis and binary exploitation, we anticipate it would be a challenge attacking the ICS systems since none of us have prior experience with such systems. We hope to leverage the knowledge and experience of our technical directors, along with our own research to explore various attack vectors and provide possible ways to detect and defend against such exploits.

One of the major issues we ran into was attempting to successfully execute a buffer overflow attack. We were unable to accomplish this due to the black box nature of the simulated environment where there was a lack of insight into what memory was being overwritten, or if the program was even taking in our full input. This led us to resort to other methods of attack to accomplish the goal of gaining remote code execution in the simulated environment.

1.7 Broader Impact

Mapping the exploits to the MITRE ATT&CK for ICS Matrix will aid in understanding the most commonly used techniques or possibly find a new technique that can be used by adversaries against ICS systems. This data can be leveraged by MITRE to build a more robust model of such adversaries and provide ways for detecting and defending against real world systems.

2. Literature Review

2.1 Executive Summary

The literature review will summarize the previous work that discusses cyber attacks on critical infrastructure. We found that industrial control systems, sometimes referred to as critical infrastructure or SCADA systems, are both incredibly important to the safety, wellbeing, and maintenance of an industrialized country, and also vulnerable to cyber attacks. There have been a number of previous cyber attacks on infrastructure that show the vulnerabilities that exist and the damage that can be caused. We also found that the nature of SCADA-based systems leaves them vulnerable. They are usually expected to be secure through other means such as security through obscurity, or secure because they are not directly connected to the internet. So the engineers that develop these systems do not take security into account. They are also often expected to be configured once and never touched again for 10-20 years, which means there are often known vulnerabilities that can be exploited. Previous research and literature noted that there has not been enough broad research that identifies an end-to-end attack pattern, rather most is focused on one attack or a specific historical attack.

2.2 Previous Work

Attacks Against Critical Infrastructure: A Global Concern - Symantec

This paper explores a handful of high-profile cyber attacks on Critical National Infrastructure (CNI), a sector that encompasses many different industries, and the methodologies used to enact the cyber attacks. In order to combat these attacks, the United States enacted The Executive Order on Improving the nation's Cybersecurity. This order called for improving software supply chain security, establishing a cyber security safety review board, creating a standard playbook for responding to cyber incidents, improving investigative and remediation capabilities, and many more.

An Undirected Attack Against Critical Infrastructure - CISA

This paper by the Department of Homeland Security outlines a few case studies for improving the control system security for critical infrastructure. Although antivirus softwares is constantly being innovated, there are still many entry points for attackers to infect their victims. Symptoms of infection include: the degradation of network performance and several other systems that make up the network. The ISP is unable to detect the compromise allowing a worm outbreak to spread throughout a facility's network. The paper discusses possible counter

measures to defend these entry points, and if they are compromised, recover as much information as possible.

Mayhem, Murder, and Misdirection: Violent Extremist Attack Plots Against Critical Infrastructure in the United States, 2016-2022 - NCITE

The primary focus of this paper is the number of extremists that threaten the United States' critical infrastructure. The paper reviews 35 cases of individuals charged in the United States federal court system that attempted to attack critical infrastructure systems. 19 of the 35 cases are linked with Salafi-jihadist activities, while 16 are associated with white supremacists.

Cyber-attack detection in SCADA systems using temporal pattern recognition techniques

This paper describes the nature of how SCADA systems are composed, which is: a sensor that collects raw data from system inputs, a programmable logic controller, and a remote telemetry unit. SCADA systems were initially secured by security through obscurity and removing their connections to the internet. Since these systems are not connected to the internet directly, but are rather connected to the infrastructure and controls the hardware that measures industrial systems, they were originally believed to be safe from internet based attacks. These systems are also vulnerable due to their long term lifespan which usually is over a decade, with no internet connectivity to release security updates. This paper suggests using an artificial intelligence model to secure these systems. The way this is suggested is through a temporal pattern recognition model. Since these systems repeat themselves in a well defined way, their behavior can be modeled using historical data across time. The results of this experiment showed that they were able to reduce attacks, however that time duration of attacks were able to evade their detection schemes.

Extending the cyber-attack landscape for SCADA-based critical infrastructure

This article from the International Journal of Critical Infrastructure Protection describes the new challenges of protecting SCADA-based critical infrastructure due to new cyber threats. They mention that the research is limited to IT attacks and protocol-specific attack demonstrations, but rarely to the big picture of attacks on SCADA infrastructure. There is thus not an understanding of the full scope that an attack on critical infrastructure might contain. The paper identifies a framework of attacks against SCADA-based infrastructure. They identified four types of attacks: traditional IT based attacks, protocol specific attacks, configuration-based attacks, and control process attacks. Protocol specific attacks involve attacks that are built to target the protocols that are used within SCADA systems, such as TCP/IP or Modbus, which is

used to serialize data to programmable logic controllers. They also found that some attacks may depend on the successful execution of other attacks.

2.3 Case Studies

The recent attack on Colonial Pipeline, which is the largest fuel pipeline in the U.S. The company's system transports roughly 2.5 million barrels of fuel daily from the Gulf Coast to the Eastern Seaboard. Attackers were able to gain remote access into Colonial Pipeline's network via a Virtual Private Network (VPN) account. The network did not use multi-factor Authentication (MFA) which is a basic and commonly used technique to authenticate the integrity of a user. This allowed the attackers to breach the network using only a compromised username and password. The attackers demanded a ransom and shut down the pipeline which had devastating impacts not only in the U.S but also globally, leading to higher fuel prices and long lines at the gas station.

In 2015, The Ukrainian power grid was hacked, resulting in power outages for approximately 230,000 Ukrainians. The cyber attack is suspected to have been executed by Russia. Russia was familiar with the software that was being used in the industrial control systems of the Ukrainian power stations, and was able to exploit the vulnerabilities. The attack was complex but consisted of spearphished emails to the corporate networks of the powerstation, utilizing BlackEnergy malware. Disabling IT infrastructure that was used to monitor the corporate network. They were able to gain control over SCADA systems and shut them down.

3. Progress Report (Final)

Our first step was configuring and setting up the project infrastructure. This involved cloning the GRFICSv3 (Github repository) and configuring each of the Virtual Machines (VMs) needed to simulate an Industrial Control System. We utilized VirtualBox and created a Host-Only Network to which each of the VMs are connected. The following figure shows the address of each of the VMs along with the Host-Only Network address.

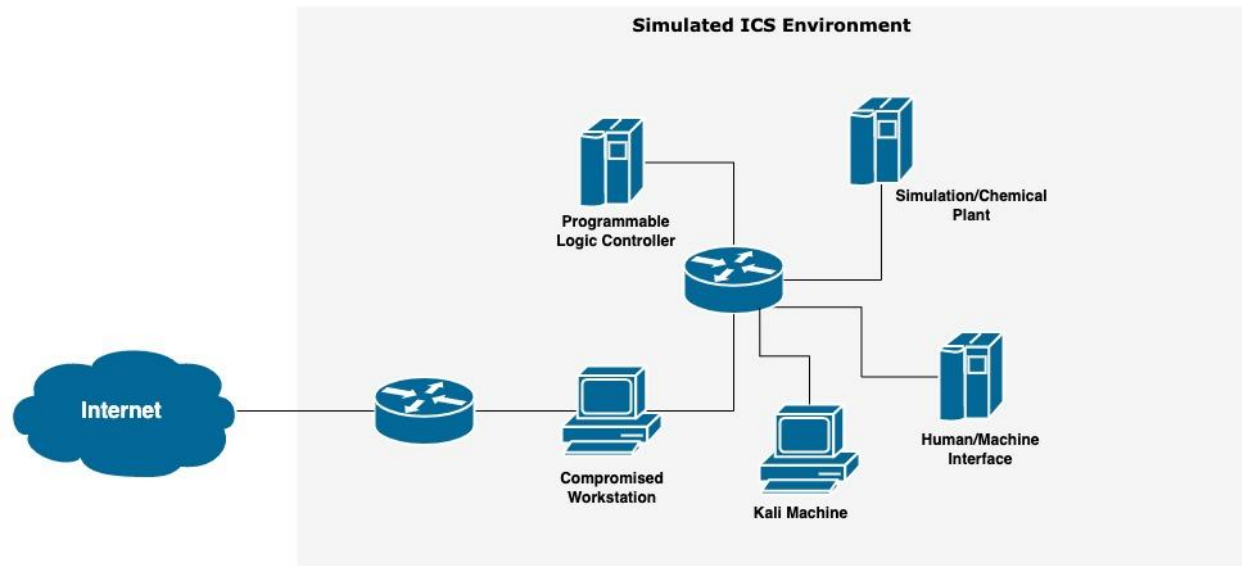


Figure: Project Infrastructure Setup

After configuration of the VMs, we ran into another issue. We had initially planned on setting up the project infrastructure on our local machines, but this proved to be inefficient as we were unable to collaborate together. We overcame this challenge by setting up a Homelab along with an RDP client to allow us to work together. Although RDP only allows for a single session at a time, we were at least able to work on the same project and each member had access to the same project documentation. Next, we planned our approach to tackle the project with the following methodology:

- Setup and Reconnaissance
- Execute a secret program
- Obtain Remote Control Access to PLC
- Access PLC's Web Interface
- Blow up Boiler

Figure: Methodology to tackle the project

3.1 Setup and Reconnaissance

Our attack scenario assumes that the attacker has access to the compromised workstation and this is where they would launch the attack from. First, the attacker would need to gain knowledge of the network and the devices/systems that are communicating on it. This step is termed Setup and Reconnaissance also referred to as Information Gathering. We used **Nmap** to scan the network along with any services and open ports running on the systems.

```
workstation@workstation:~$ nmap -A 192.168.95.111/24
Starting Nmap 7.80 ( https://nmap.org ) at 2022-10-20 14:44 EDT
Nmap scan report for 192.168.95.2
Host is up (0.00073s latency).
Not shown: 999 closed ports
PORT      STATE SERVICE VERSION
22/tcp    open  ssh      OpenSSH 7.2p2 Ubuntu 4ubuntu2.4 (Ubuntu Linux; protocol 2.0)
|_ ssh-hostkey:
|_ 2048 19:c7:93:27:d7:fa:95:3c:d1:0e:ic9:dd:9b:1f:f4:cf (RSA)
|_ 256 de:bc:c6:2d:2d:76:b2:b1:49:eb:c8:cf:01:4b:f5:40 (ECDSA)
|_ 256 65:d6:e5:68:f2:be:ad:3e:0b:b3:44:4e:c0:28:26:ff (ED25519)
Service Info: OS: Linux; CPE: cpe:/o:linux:linux_kernel

Nmap scan report for 192.168.95.5
Host is up (0.00087s latency).
Not shown: 997 closed ports
PORT      STATE SERVICE VERSION
22/tcp    open  ssh      OpenSSH 7.4p1 Debian 10+deb9u7 (protocol 2.0)
|_ ssh-hostkey:
|_ 2048 c7:a8:4b:8a:9b:1a:20:60:47:1e:a8:e2:e1:fb:25:9d (RSA)
|_ 256 53:92:f9:58:fc:83:93:ef:d4:14:bf:9e:00:2f:b4:a9 (ECDSA)
|_ 256 8a:62:75:20:c6:e5:fb:45:99:d4:4d:b9:45:5a:10:99 (ED25519)
8009/tcp  open  ajp13    Apache Jserv (Protocol v1.3)
|_ ajp-methods:
|_ Supported methods: GET HEAD POST PUT DELETE OPTIONS
|_ Potentially risky methods: PUT DELETE
|_ See https://nmap.org/nsedoc/scripts/ajp-methods.html
8080/tcp  open  http     Apache Tomcat/Coyote JSP engine 1.1
|_ http-favicon: Apache Tomcat
|_ http-methods:
|_ Potentially risky methods: PUT DELETE
|_ http-open-proxy: Proxy might be redirecting requests
|_ http-server-header: Apache-Coyote/1.1
|_ http-title: Apache Tomcat
Service Info: OS: Linux; CPE: cpe:/o:linux:linux_kernel

Nmap scan report for Workstation (192.168.95.6)
Host is up (0.00069s latency).
Not shown: 999 closed ports
PORT      STATE SERVICE VERSION
22/tcp    open  ssh      OpenSSH 8.2p1 Ubuntu 4ubuntu0.1 (Ubuntu Linux; protocol 2.0)
Service Info: OS: Linux; CPE: cpe:/o:linux:linux_kernel

Nmap scan report for 192.168.95.10
Host is up (0.00070s latency).
Not shown: 998 closed ports
PORT      STATE SERVICE VERSION
22/tcp    open  ssh      OpenSSH 7.2p2 Ubuntu 4ubuntu2.7 (Ubuntu Linux; protocol 2.0)
|_ ssh-hostkey:
|_ 2048 91:b4:38:6f:5e:72:a4:c8:ad:7c:64:56:50:9e:ce:63 (RSA)
|_ 256 91:c7:b1:20:b7:35:51:9a:8d:ca:b7:86:c9:81:e0:5c (ECDSA)
```

Figure: Nmap scan of the network 192.168.95.111/24

We were able to identify various services such as SSH and Apache Tomcat Servers running on the various machines on common ports such as 80 and 8080 but also on some unusual ones such as 55555. We examined each of the services and this led us to an administrative panel for the Human Machine Interface (HMI). The page required a username and password which were found to be admin and admin respectively using a brute force attack with a common list of usernames and passwords. This demonstrated our first vulnerability of **Weak Default Credentials**, in which network administrators forget to change the weak default credentials supplied by the software manufacturers. This allows attackers to easily gain access to sensitive pages which otherwise they could not.

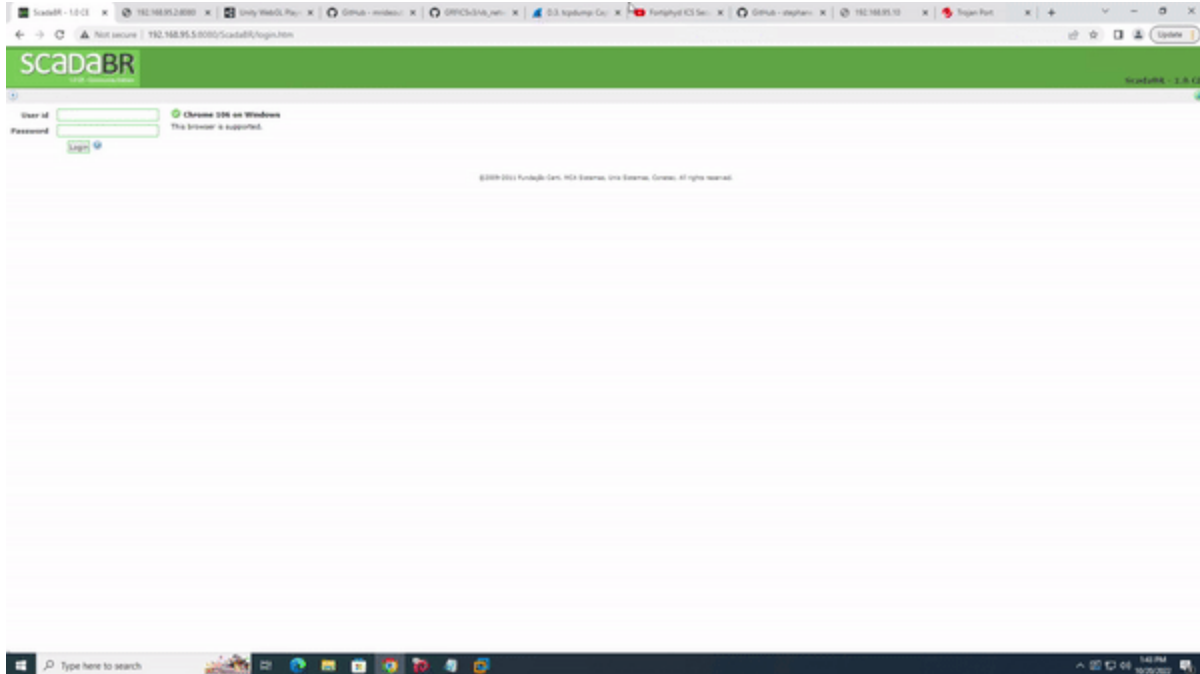
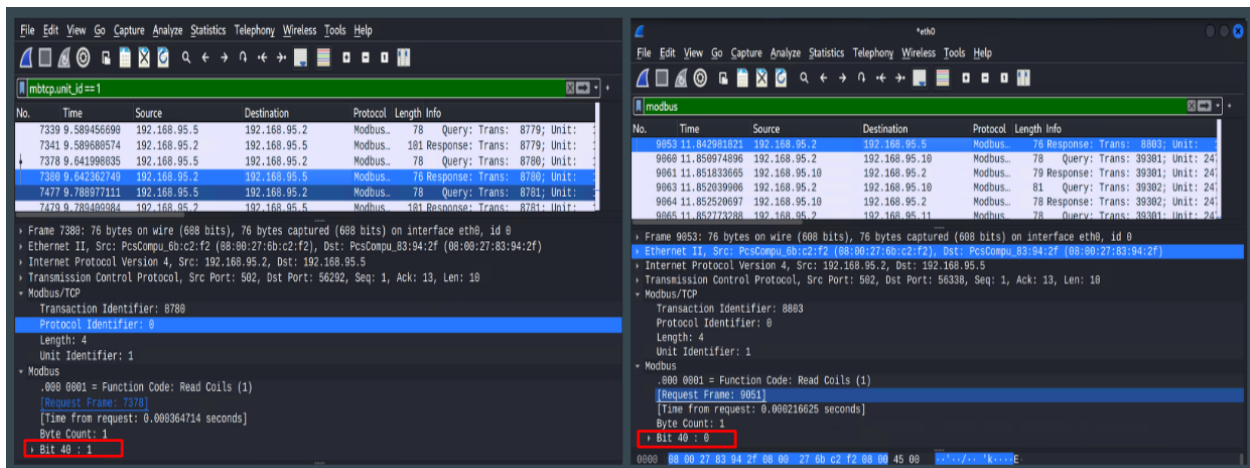


Figure: Weak Default Credentials on Human Machine Interface

3.2 Network Sniffing

We performed a Wireshark packet capture of the Modbus communication between the PLC and the HMI. We were able to determine the bit state of the communication to the HMI by analyzing the packets. A bit value of 1 corresponds to START and a bit value of 0 corresponds to STOP



3.3 Change Operating Mode

We were able to perform the Change of Operating Mode (T0858) attack via Unauthorized Modbus Command Injection



3.4 Malicious Program Upload

This attack aims to blow up the boiler on the chemical plant by uploading a malicious program to demonstrate the severity an attacker could cause in real world systems.



3.5 Brute Forcing SSH Credentials

We were able to bruteforce the SSH credentials of the PLC by running Hydra, a brute forcing tool on Kali Linux. We used a wordlist containing commonly used usernames and passwords.

```
(kali@kali)-[~/Downloads]
$ hydra -l default-usenames.txt -P default-passwords.txt 192.168.95.2 ssh -t 5
Hydra v9.3 (c) 2022 by van Hauser/THC & David Maciejak - Please do not use in military or secret service organizations, or for illegal purposes (this is non-binding, these *** ignore laws and ethics anyway).

Hydra (https://github.com/vanhauser-thc/thc-hydra) starting at 2022-11-17 19:01:26
[DATA] max 5 tasks per 1 server, overall 5 tasks, 238 login tries (l:17/p:14), ~48 tries per task
[DATA] attacking ssh://192.168.95.2:22/
[22][ssh] host: 192.168.95.2 login: user password: password
[STATUS] 95.00 tries/min, 95 tries in 00:01h, 143 to do in 00:02h, 5 active
[STATUS] 73.00 tries/min, 146 tries in 00:02h, 92 to do in 00:02h, 5 active
[STATUS] 70.67 tries/min, 212 tries in 00:03h, 26 to do in 00:01h, 5 active
1 of 1 target successfully completed, 1 valid password found
Hydra (https://github.com/vanhauser-thc/thc-hydra) finished at 2022-11-17 19:04:40
```

```
workstation@Workstation:~$ ssh -l user 192.168.95.2
user@192.168.95.2's password:
Welcome to Ubuntu 16.04.4 LTS (GNU/Linux 4.4.0-116-generic i686)

 * Documentation:  https://help.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/advantage

246 packages can be updated.
163 updates are security updates.

Last login: Wed Nov 30 01:01:19 2022
user@plc:~$ id
uid=1000(user) gid=1000(user) groups=1000(user),4(adm),24(cdrom),27(sudo),30(dip),46(plugdev),110(lxd),115(lpadmin),116(sambashare)
user@plc:~$
```

3.6 Upload Malicious Program

```
FUNCTION_BLOCK pressure_control
VAR
    pressure_real : REAL := 2700.0;
END_VAR
VAR_INPUT
    pressure : UINT := 58981;
END_VAR
VAR
    pressure_sp_real : REAL := 2700.0;
END_VAR
VAR_INPUT
    pressure_sp : UINT := 65535;
    curr_pos : UINT := 30000;
END_VAR
VAR
    valve_pos_real : REAL := 39.25;
    pos_update_real : REAL := 0.0;
    valve_pos_nominal : REAL := 39.25;
END_VAR
VAR_OUTPUT
    valve_pos : UINT := 25886;
END_VAR
```

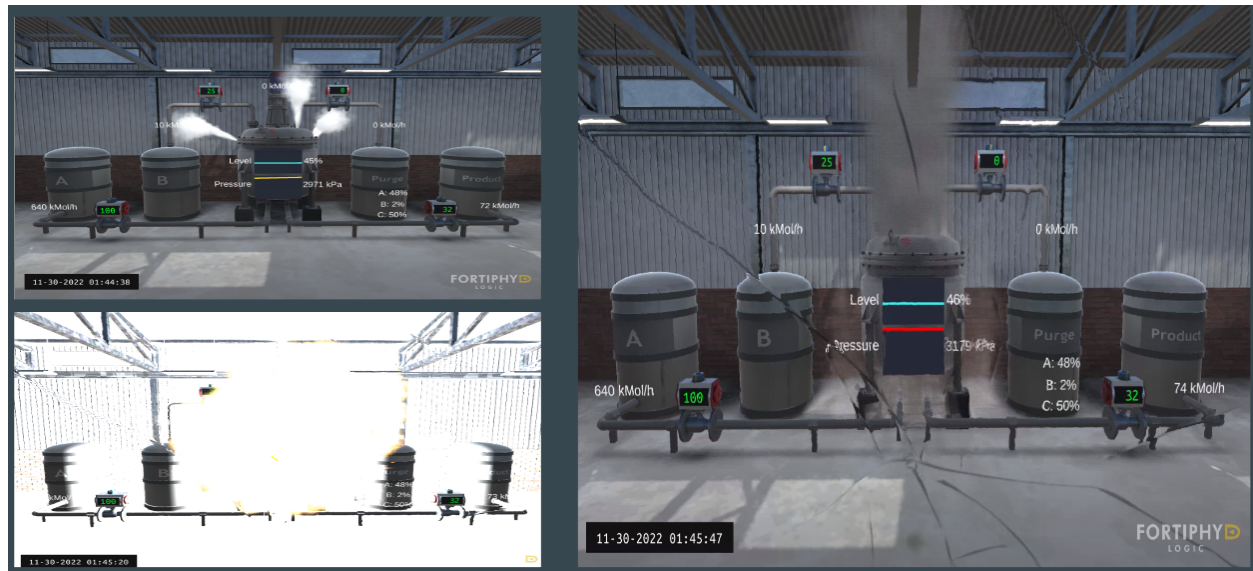
```
workstation@Workstation:~/Desktop$ scp attack.st user@192.168.95.2:/home/user/
attack.st
workstation@Workstation:~/Desktop$ ssh -l user 192.168.95.2
user@192.168.95.2's password:
Welcome to Ubuntu 16.04.4 LTS (GNU/Linux 4.4.0-116-generic i686)

 * Documentation:  https://help.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/advantage

246 packages can be updated.
163 updates are security updates.

Last login: Wed Nov 30 02:28:02 2022 from 192.168.95.6
user@plc:~$ ls
attack.st dump.pcap OpenPLC_v2 oplc.tar.gz st_files
user@plc:~$
```


3.7 Blowing Up The Boiler



3.8 Mapping the attacks to the MITRE ATT&CK for ICS Matrix

Initial Access	Execution	Persistence	Privilege Escalation	Evasion	Discovery	Lateral Movement	Collection	Command and Control	Inhibit Response Function	Impair Process Control	Impact
Drive-by Compromise	Change Operating Mode	Modify Program	Exploitation for Privilege Escalation	Change Operating Mode	Network Connection Enumeration	Default Credentials	Automated Collection	Commonly Used Port	Activate Firmware Update Mode	Brute Force I/O	Damage to Property
Exploit Public-Facing Application	Command-Line Interface	Module Firmware	Hooking	Exploitation for Evasion	Network Sniffing	Exploitation of Remote Services	Data from Information Repositories	Connection Proxy	Alarm Suppression	Modify Parameter	Denial of Control
Exploitation of Remote Services	Execution through API	Project File Infection		Indicator Removal on Host	Remote System Discovery	Lateral Tool Transfer	Detect Operating Mode	Standard Application Layer Protocol	Block Command Message	Module Firmware	Denial of View
External Remote Services	Graphical User Interface	System Firmware		Masquerading	Remote System Information Discovery	Program Download	I/O Image		Block Reporting Message	Spoof Reporting Message	Loss of Availability
Internet Accessible Device	Hooking	Valid Accounts		Rootkit	Wireless Sniffing	Remote Services	Man in the Middle		Block Serial COM	Unauthorized Command Message	Loss of Control
Remote Services	Modify Controller Tasking			Spoof Reporting Message		Valid Accounts	Monitor Process State		Data Destruction		Loss of Productivity and Revenue
Replication Through Removable Media	Native API						Point & Tag Identification		Denial of Service		Loss of Protection
Rogue Master	Scripting						Program Upload		Device Restart/Shutdown		Loss of Safety
Spearphishing Attachment	User Execution						Screen Capture		Manipulate I/O Image		Loss of View
Supply Chain Compromise							Wireless Sniffing		Modify Alarm Settings		Manipulation of Control
Transient Cyber Asset									Rootkit		Manipulation of View
Wireless Compromise									Service Stop		Theft of Operational Information
									System Firmware		

Figure: MITRE ATT&CK for ICS Matrix

Based on the attacks we have conducted so far we were able to map them to the following codes in the ICS Matrix:

Tactics for Modbus Command Injection

Based on MITRE ICS Matrix our exploits covered the following areas:

- Remote Services - T0886
- Change Operating Mode - T0858
- Network Connection Enumeration - T0840
- Network Sniffing - T0842
- Default Credentials - T0812
- Valid Account - T0859
- Loss of Productivity and Revenue - T0828
- Manipulation of Control - T0831

Tactics for Malicious Program Upload

Based on MITRE ICS Matrix our exploits covered the following areas:

- Remote Services - T0886
- Network Connection Enumeration - T0840
- Default Credentials - T0812
- Valid Accounts - T0859
- Program Upload - T0845
- Damage to Property - T0879

4. Future Plans

4.1 Use a buffer overflow attack to gain remote code execution

One of our goals that we were unable to accomplish due to lack of available resources, which led us to not complete that goal, and we resorted to other methods of attack. This is not unrealistic of a real world hacking attempt where multiple vectors of attack may be attempted for only a few to succeed.

5. Background of Students

Kushan Singh is a second-year Computer Science graduate student at The University of Texas at Dallas with a Bachelor's in Information Science and Engineering from India. He has worked as a Backend Developer managing and creating user authentication and access controls, maintaining the database and storing secure passwords, and building a REST API. In his most recent internship as an Application Security Analyst he has gained experience in penetration testing web application, code reviews, static and dynamic code analysis, and secure code practices. He has taken various security courses as part of his graduate degree such as Network Security, System Security and Malware Analysis, and Data and Application Security. He is well versed in building web applications with the MERN stack, and has development experience with C, C++ and Python as well. He is familiar with the Intel x86 architecture set and has worked with tools such as GDB, Ghidra, Burp Suite and Metasploit, He has also participated in various Capture the Flag (CTF) events.

Harrison Hamric is a first-year Computer Science graduate student with a Bachelors in Computer Science from the University of Texas at Dallas. His background is working as a Software Developer and Cybersecurity Researcher for both private industry and the US Government. His most recent position was as an intern on a team of Cybersecurity Researchers, where he had to develop and analyze Windows system code. He has taken the Windows User Mode Exploit Development course from Offensive Security, and has a Security+ certification. He has taken courses in Operating Systems, several courses in C/C++ development, and is currently taking a course on Reverse Engineering and Malware Analysis. Analyzing and developing system code, as well as taking courses in reverse engineering and binary analysis has given him skills that will be relevant to the project of identifying concealment techniques for malware in a modern environment where techniques are becoming more advanced and harder to identify. He has also competed in multiple CTF events.

Hafiz Akinteye is a first-year Computer Science graduate student with a Bachelors in Computer Science from the University of Texas at Dallas. He has taken multiple relevant courses such as Operating Systems, Programming in a UNIX Environment, and Computer Networks. Currently, Hafiz is working on attaining his Security+ certification and is taking a course on Malware Analysis. He has engaged in projects such as replacing the standard round robin scheduler to a lottery scheduler in the operating system. In addition, Hafiz is experienced in Python, C++, C, and Java.

Akshay Jain is a second-year Computer Science graduate student at The University of Texas at Dallas with a Bachelor's in Computer Engineering from India. He has worked as a DevSecOps Engineer managing and creating Cyber threat detections and scoring of cyber risks

based on threat information from other external sources. He has taken various security courses as part of his graduate degree such as Network Security, Information Security, and Data and Application Security. He has development experience with C, C++ and Python as well. He has worked with tools such as GDB, Ghidra, Burp Suite and Metasploit, Snort, Wireshark, nmap etc., He has also participated in various hackathons and Capture the Flag (CTF) events.

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